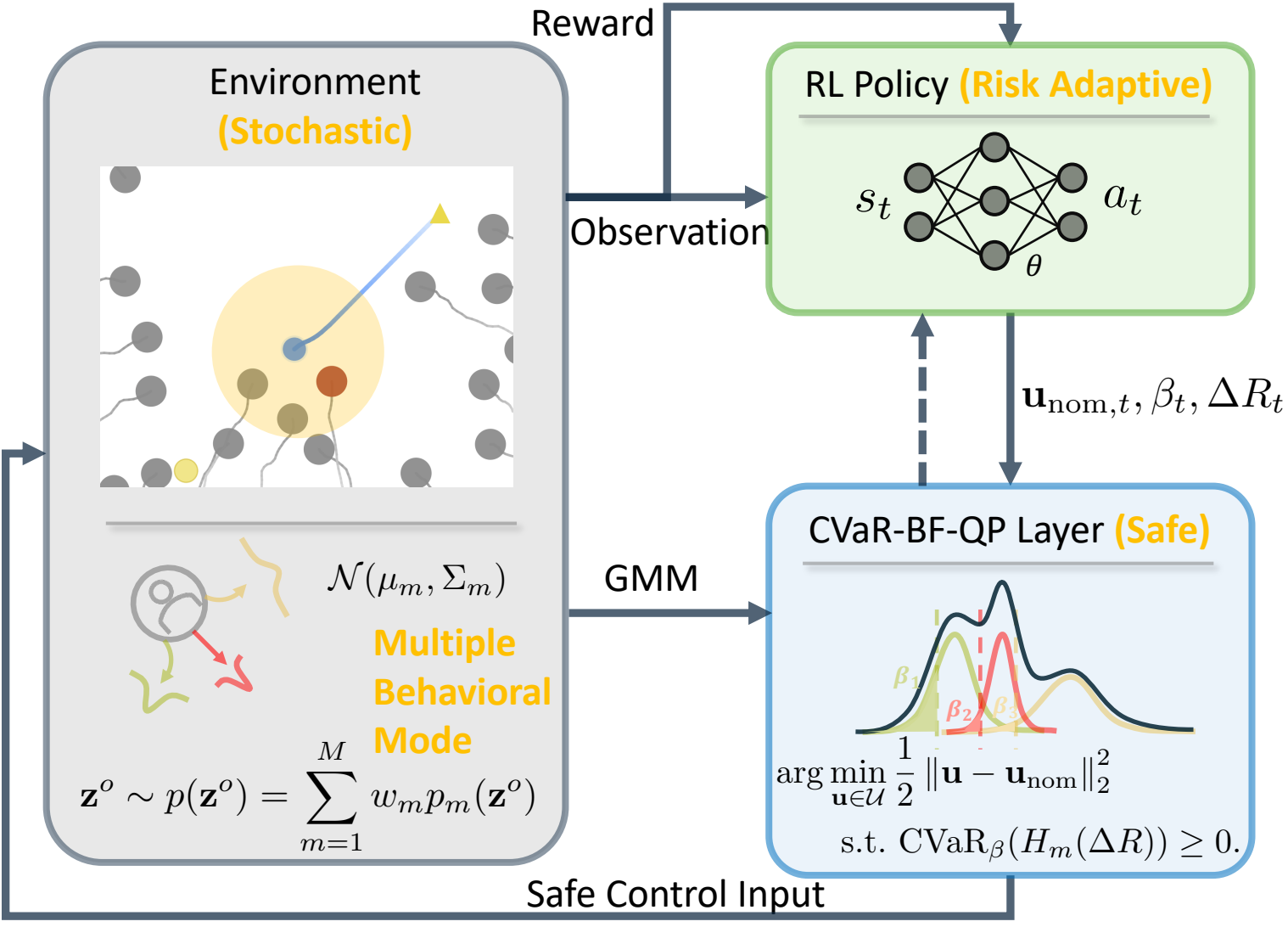
Reward
Environment
(Stochastic)
RL Policy (Risk Adaptive)
s_t
a_t
θ
Observation
$\mathbf{u}_{\mathrm{nom},t}, \beta_t, \Delta R_t$
$\mathcal{N}(\mu_m, \Sigma_m)$
Multiple
Behavioral
Mode
$\mathbf{z}^o \sim p(\mathbf{z}^o) = \sum_{m=1}^{M} w_m p_m(\mathbf{z}^o)$
GMM
CVaR-BF-QP Layer (Safe)
β_1
β_2
β_3
$\arg\min_{\mathbf{u}\in\mathcal{U}} \frac{1}{2}\|\mathbf{u}-\mathbf{u}_{\mathrm{nom}}\|_2^2$
s.t. $\mathrm{CVaR}_\beta(H_m(\Delta R)) \geq 0.$
Safe Control Input